SHAHBAZ: Custom Python Chart Generator & Dashboard Builder

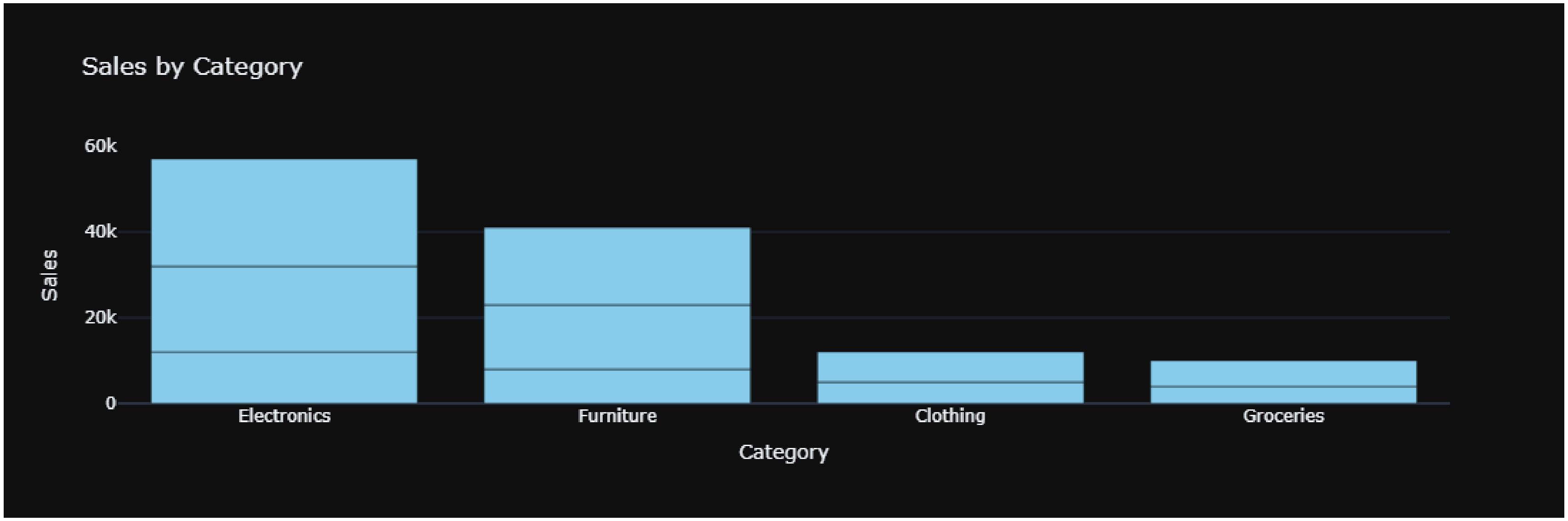
In [5]:
```
import sys
sys.path.append(r"C:\Users\USER\Downloads")
```

In [6]:
```
import SHAHBAZ as BAZ
```

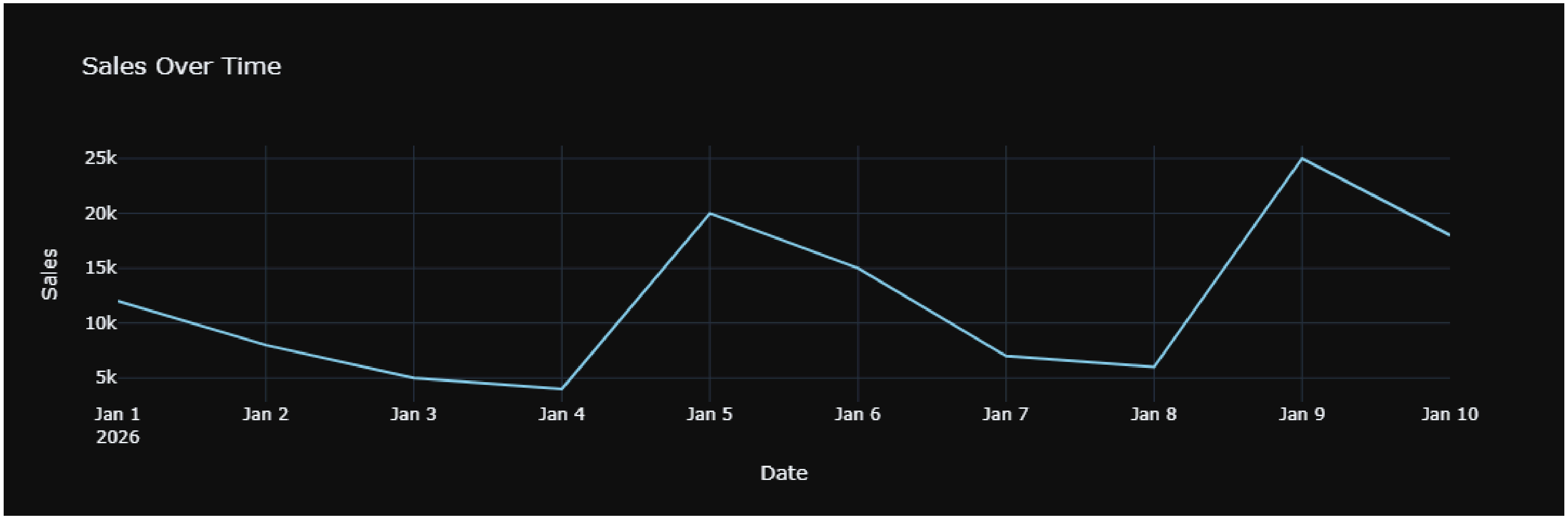
In [ ]:
```
import pandas as pd

data = {
    "Date": [
        "2026-01-01","2026-01-02","2026-01-03","2026-01-04","2026-01-05",
        "2026-01-06","2026-01-07","2026-01-08","2026-01-09","2026-01-10"
    ],
    "Region": ["North","South","East","West","North","South","East","West","North","South"],
    "Category": ["Electronics","Furniture","Clothing","Groceries","Electronics",
                 "Furniture","Clothing","Groceries","Electronics","Furniture"],
    "Product": ["Phone","Chair","Shirt","Rice","Laptop",
                "Table","Jeans","Oil","TV","Sofa"],
    "Segment": ["Retail","Wholesale","Retail","Wholesale","Retail",
                "Wholesale","Retail","Wholesale","Retail","Wholesale"],
    "Sales": [12000, 8000, 5000, 4000, 20000, 15000, 7000, 6000, 25000, 18000],
    "Profit": [3000, 2000, 1200, 900, 5000, 3500, 1500, 1300, 7000, 4000],
    "Quantity": [2, 4, 5, 8, 1, 2, 3, 6, 1, 1],
    "Rating": [4.5, 4.0, 3.8, 4.2, 4.7, 4.1, 3.9, 4.3, 4.8, 4.4]
}

df = pd.DataFrame(data)
df["Date"] = pd.to_datetime(df["Date"])
df
```

In [77]:
```
# SINGLE CHART TESTS
```

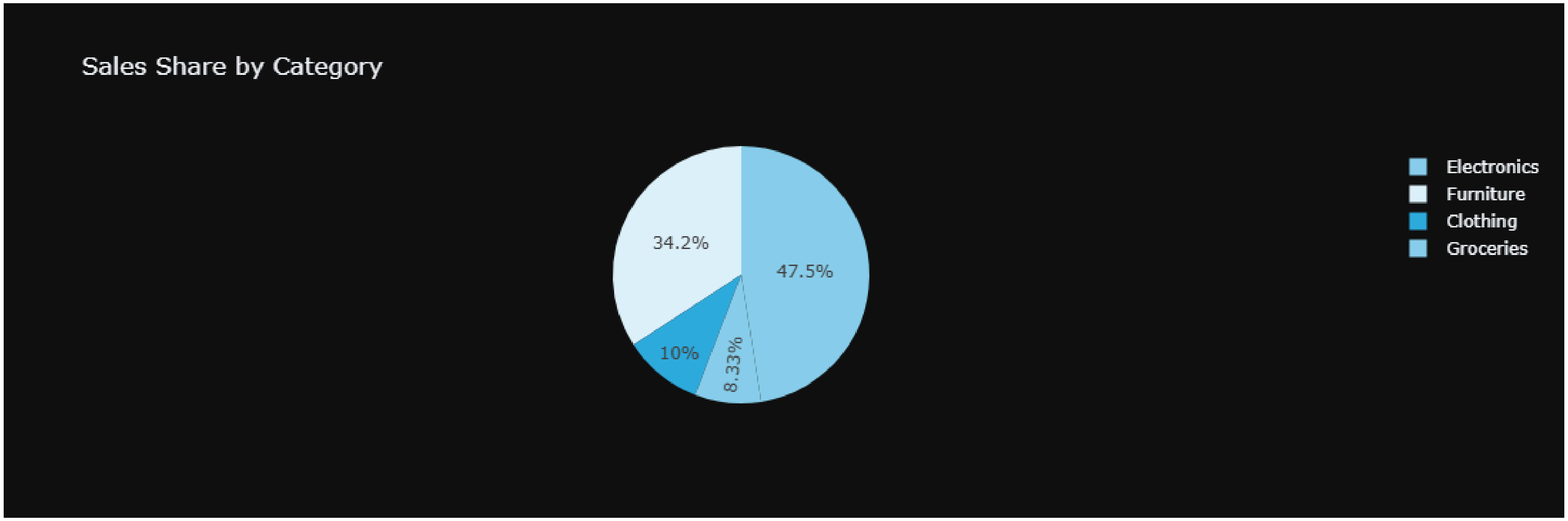
In [58]:
```
# 1. bar
BAZ.chart(df, "bar", x="Category", y="Sales", title="Sales by Category")
```



In [57]:
```python
# 2. line
BAZ.chart(df, "line", x="Date", y="Sales", title="Sales Over Time")
```

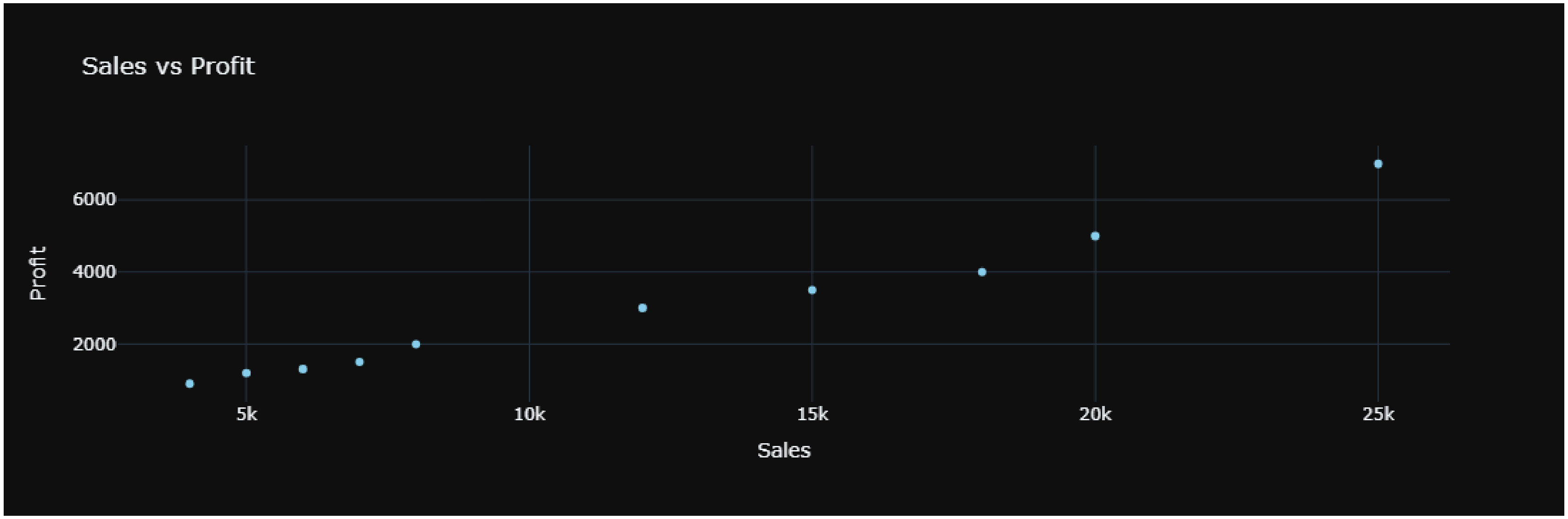


In [56]:
```python
# 3. pie
BAZ.chart(df, "pie", x="Category", y="Sales", title="Sales Share by Category")
```



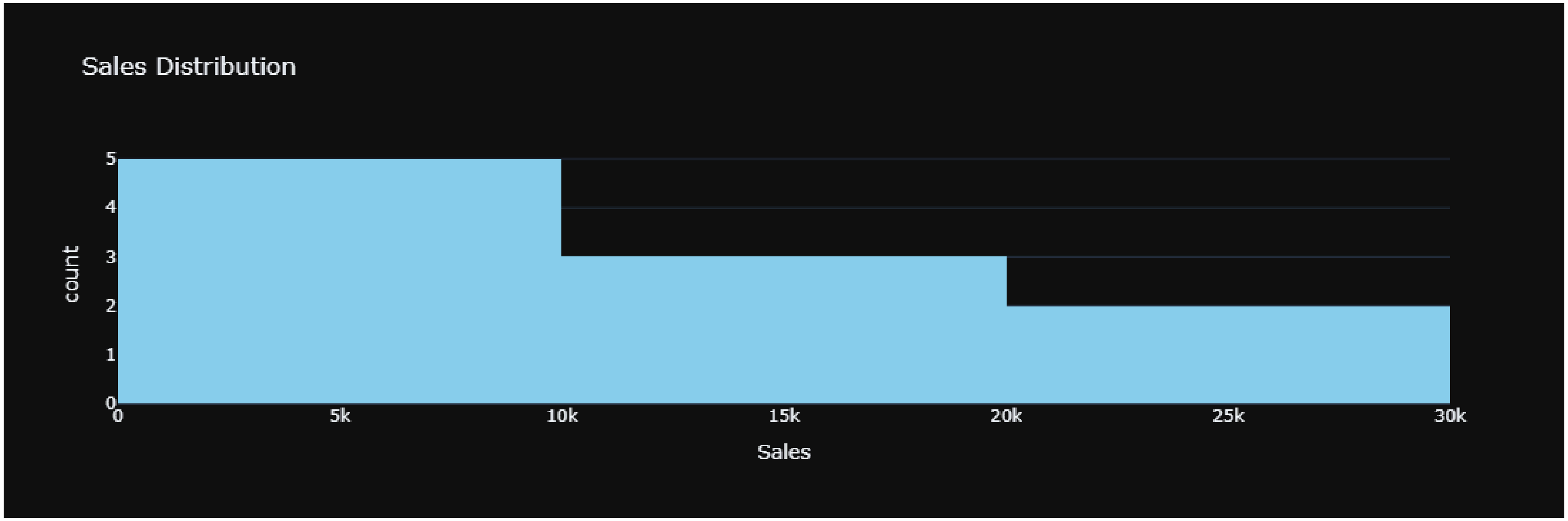
In [55]:

```
# 4. scatter
BAZ.chart(df, "scatter", x="Sales", y="Profit", title="Sales vs Profit")
```



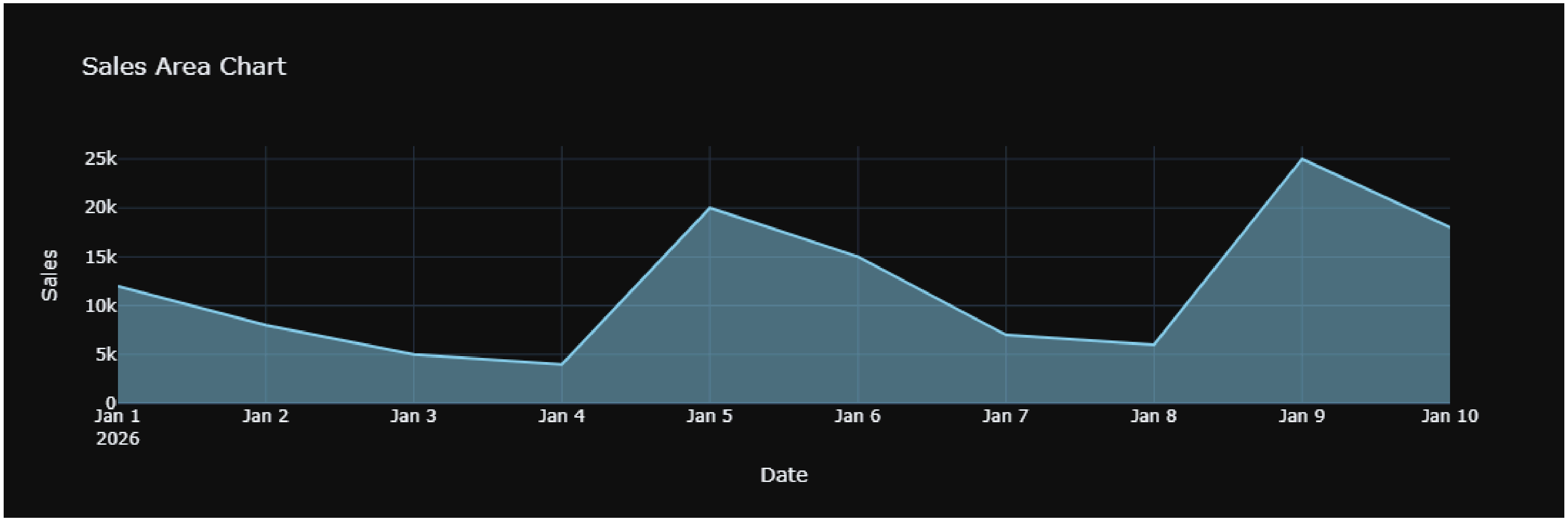
In [54]:

```
# 5. hist
BAZ.chart(df, "hist", x="Sales", title="Sales Distribution")
```



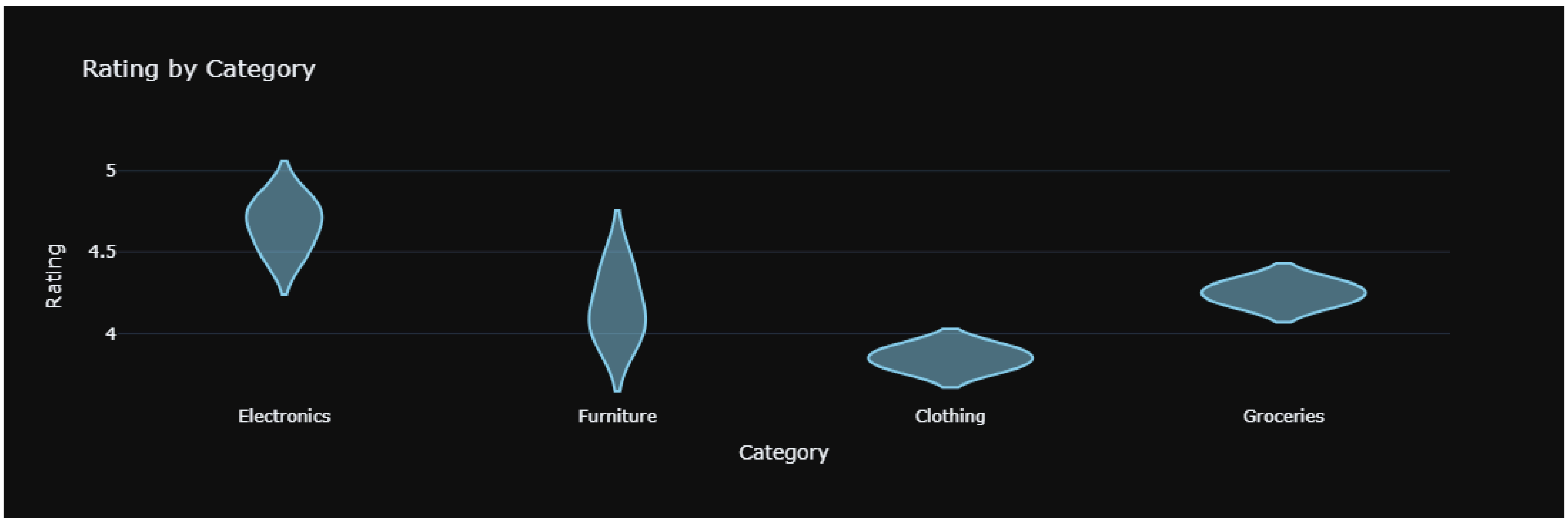
In [53]:
```
# 6. box
BAZ.chart(df, "box", x="Category", y="Profit", title="Profit by Category")
```

In [52]:
```
# 7. area
BAZ.chart(df, "area", x="Date", y="Sales", title="Sales Area Chart")
```



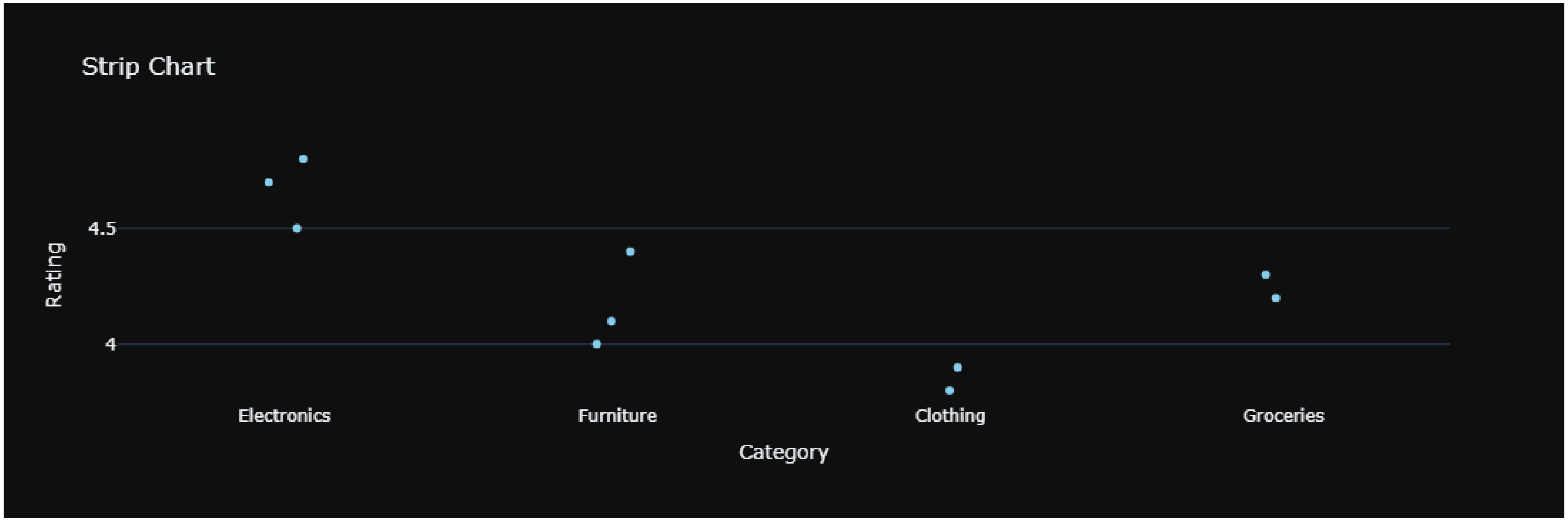
In [51]:

```
# 8. violin
BAZ.chart(df, "violin", x="Category", y="Rating", title="Rating by Category")
```

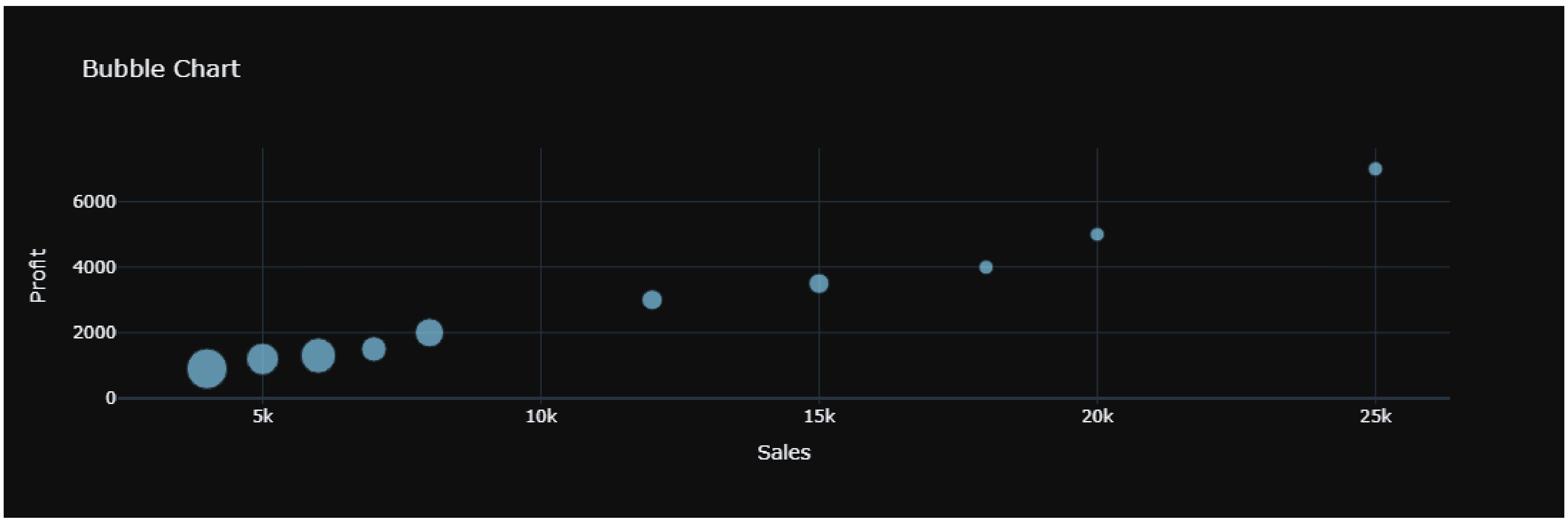


In [50]:

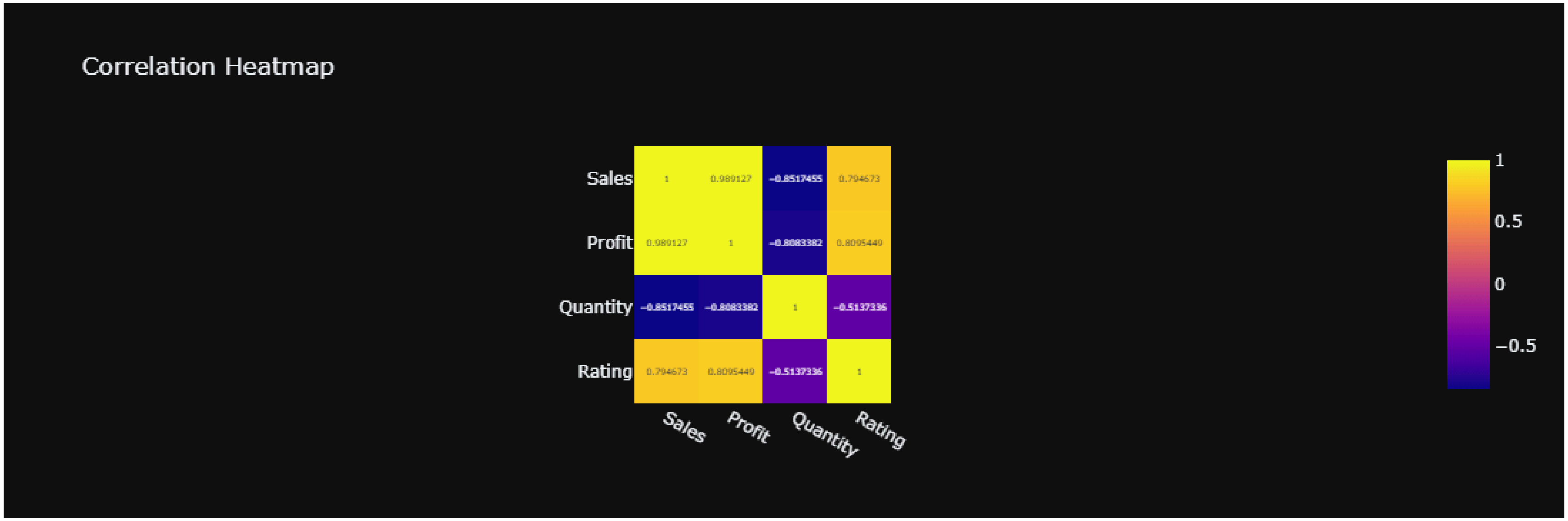
```
# 9. strip
BAZ.chart(df, "strip", x="Category", y="Rating", title="Strip Chart")
```



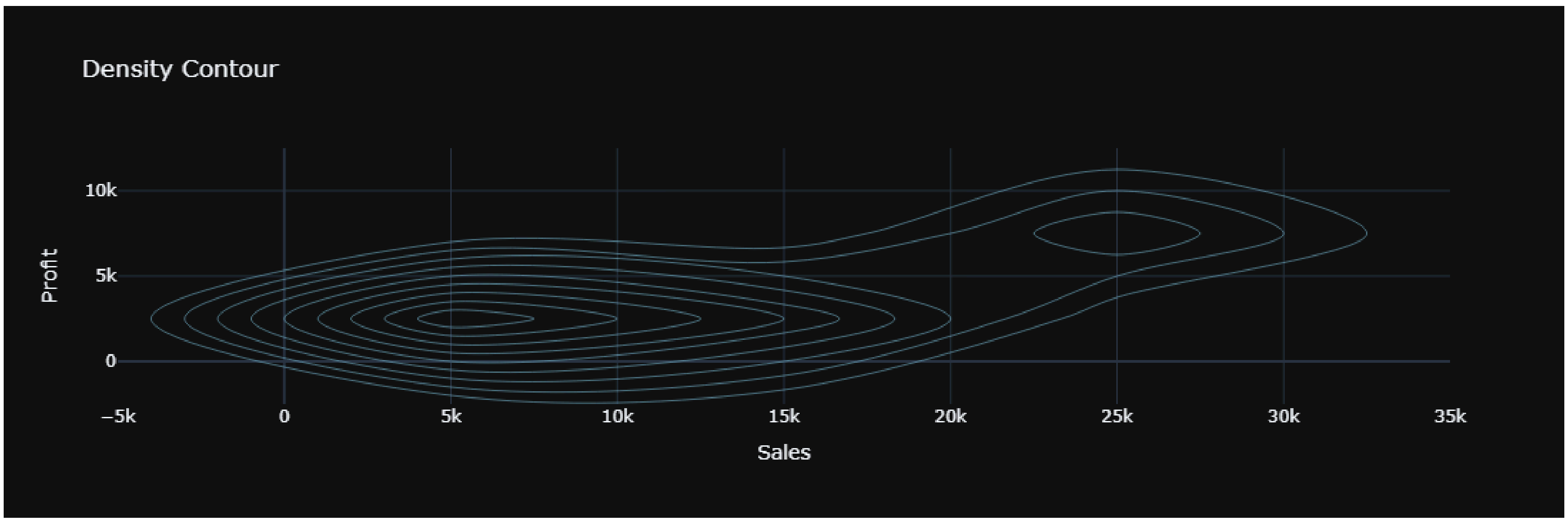
In [49]:
```
# 10. bubble
BAZ.chart(df, "bubble", x="Sales", y="Profit", size="Quantity", title="Bubble Chart")
```



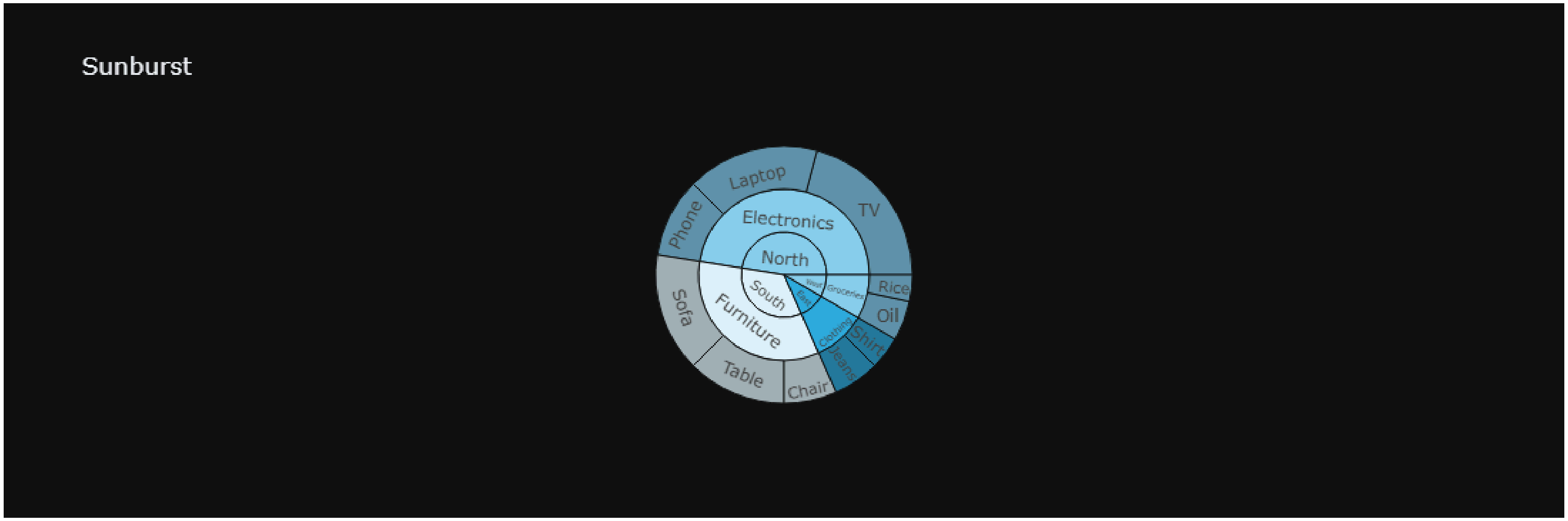
In [75]:
```
# 11. heatmap
BAZ.chart(df, "heatmap", title="Correlation Heatmap")
```



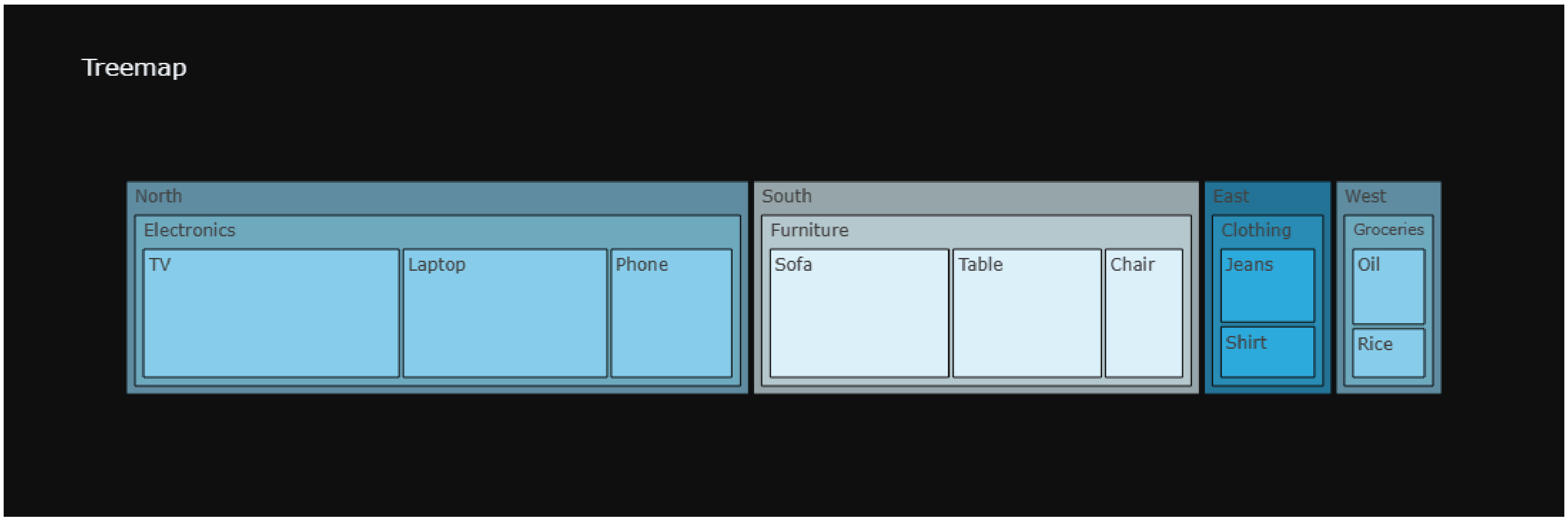
In [74]:
```python
# 13. density_contour
BAZ.chart(df, "density_contour", x="Sales", y="Profit", title="Density Contour")
```



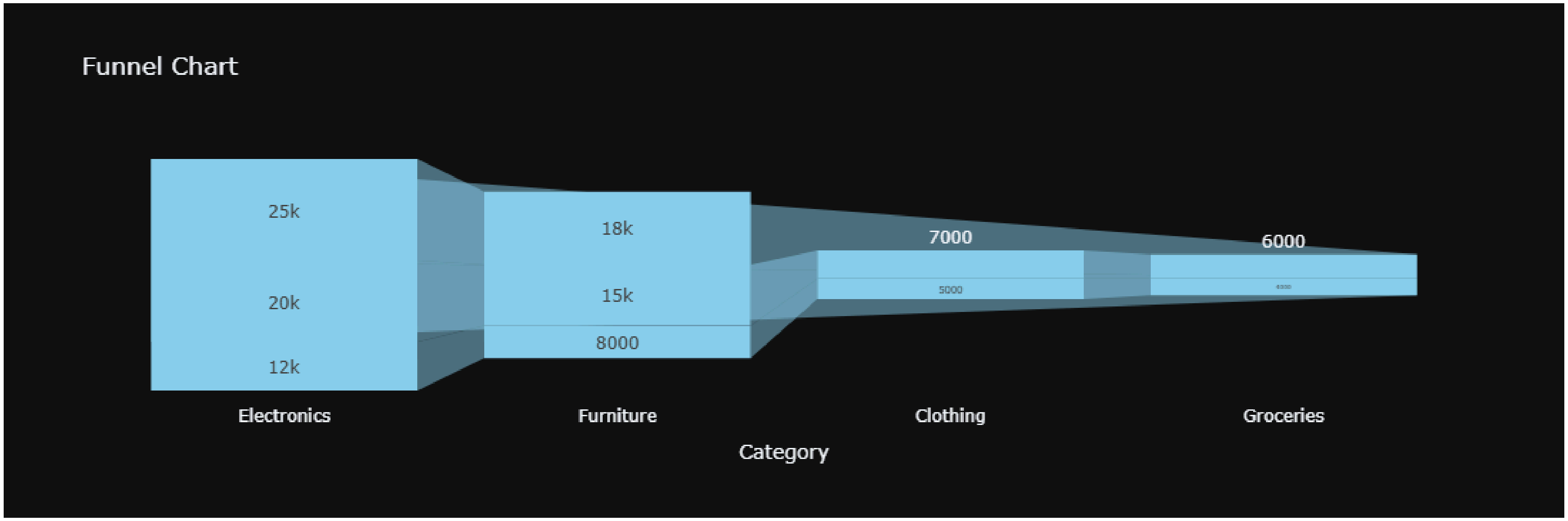
In [73]:
```python
# 14. sunburst
BAZ.chart(df, "sunburst", path=["Region", "Category", "Product"], y="Sales", title="Sunburst")
```



In [45]:
```
# 15. treemap
BAZ.chart(df, "treemap", path=["Region", "Category", "Product"], y="Sales", title="Treemap")
```

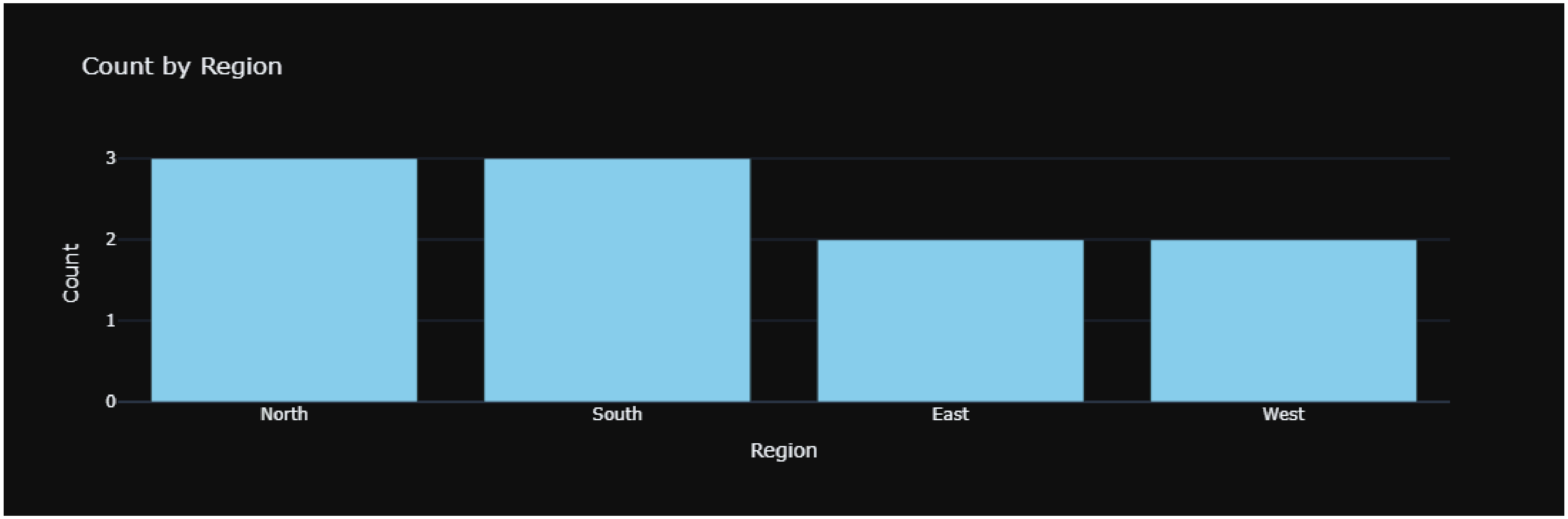


In [44]:
```
# 16. funnel
BAZ.chart(df, "funnel", x="Category", y="Sales", title="Funnel Chart")
```



In [72]:
```
# GROUPBY TESTS
```

In [43]:
```
# count by region
BAZ.chart(df, "bar", groupby_col="Region", agg_func="count", title="Count by Region")
```

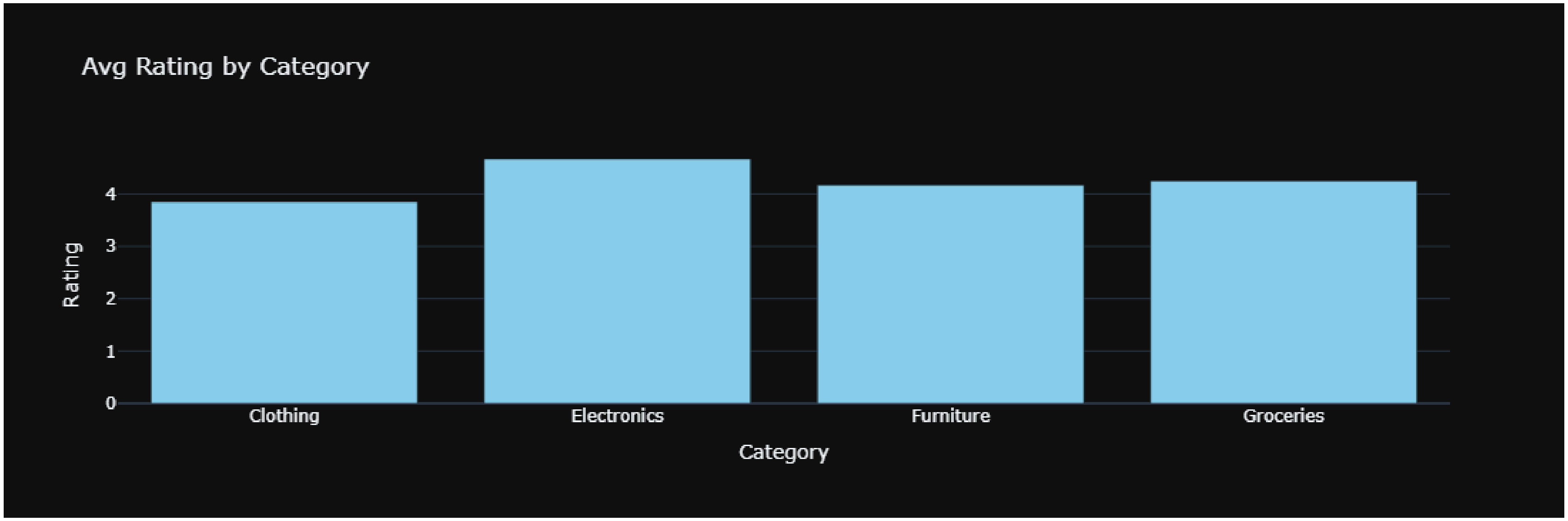


In [27]:
```
# sales by city
```

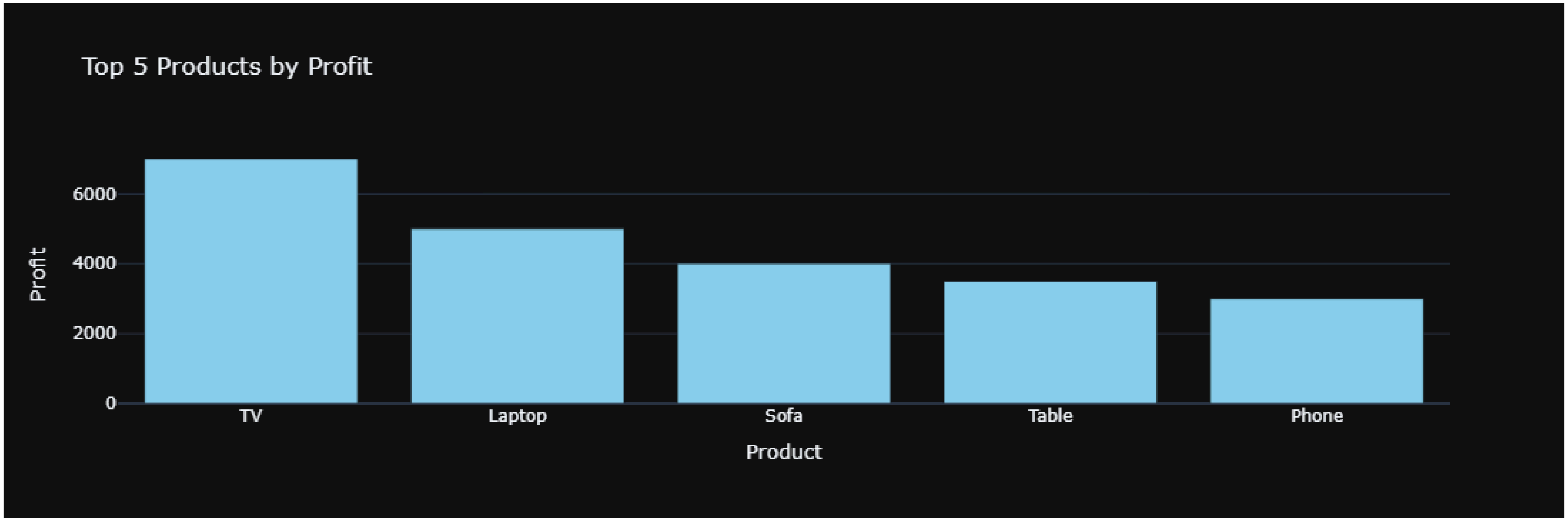
```
BAZ.chart(df, "bar", groupby_col="City", agg_col="Sales", agg_func="sum", title="Sales by City")
```

groupby_col "City" not found in dataframe columns

In [42]:
```
# avg rating by category
BAZ.chart(df, "bar", groupby_col="Category", agg_col="Rating", agg_func="mean", title="Avg Rating by Category")
```

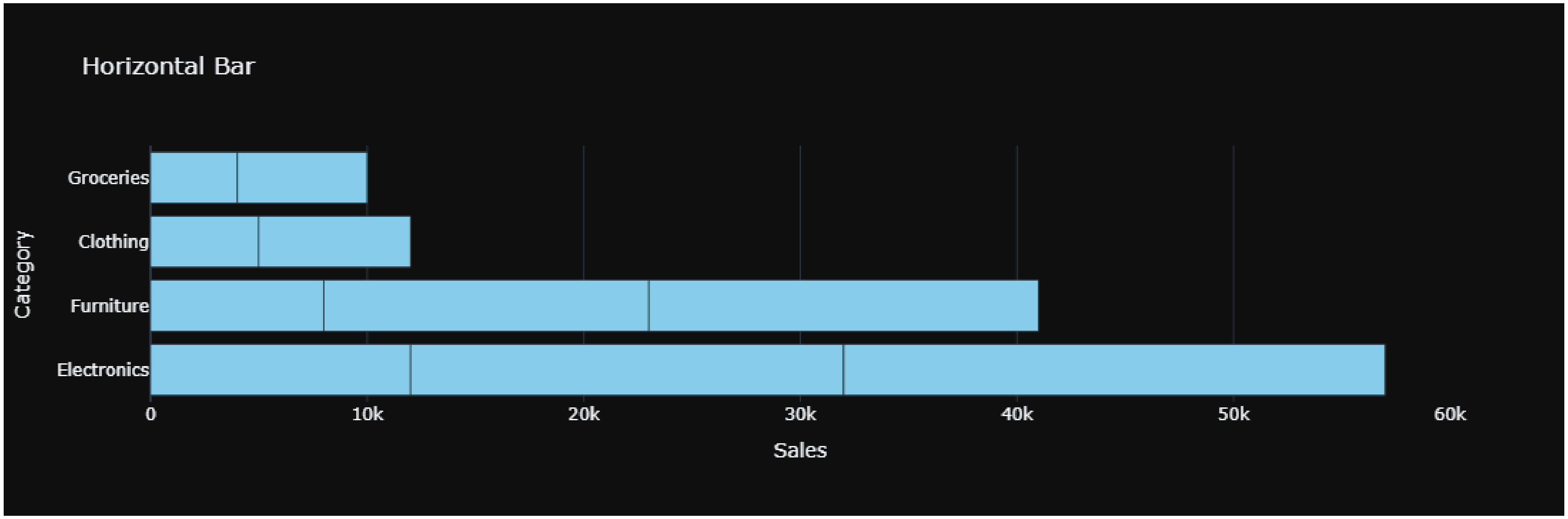


In [41]:
```
# top 5 products by profit
BAZ.chart(df, "bar", groupby_col="Product", agg_col="Profit", agg_func="sum",
          sort=True, top_n=5, title="Top 5 Products by Profit")
```



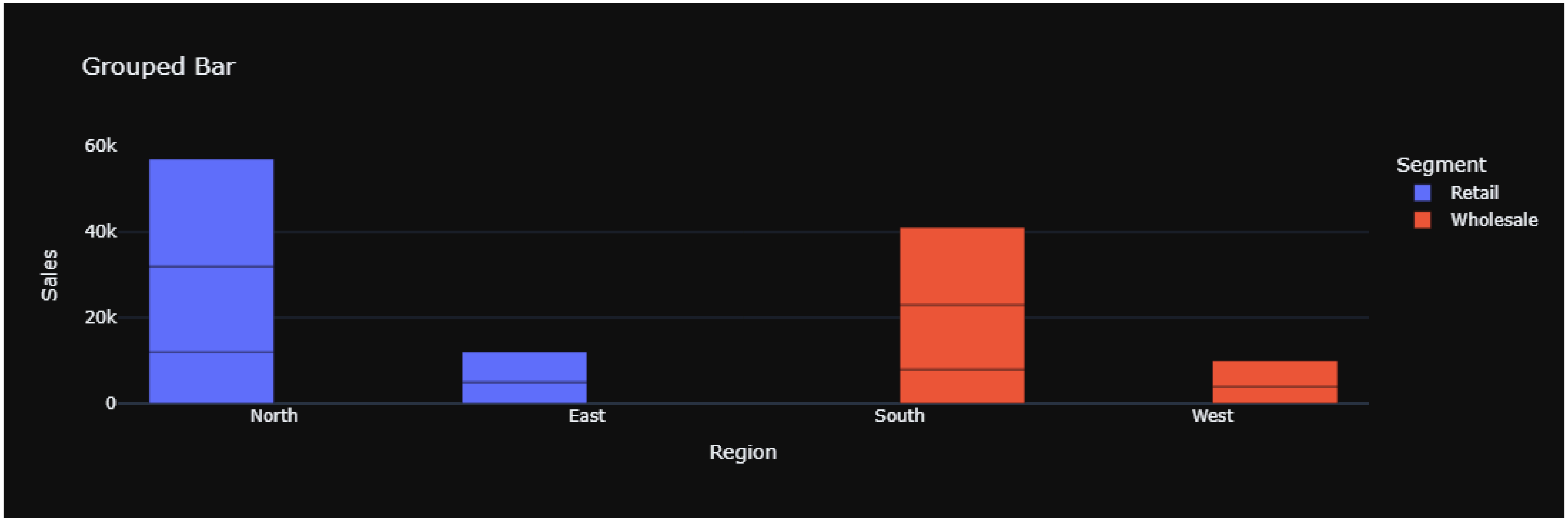
In [30]:
```
# BAR OPTIONS TESTS
```

In [40]:
```
# horizontal bar
BAZ.chart(df, "bar", x="Category", y="Sales", orientation="h", title="Horizontal Bar")
```



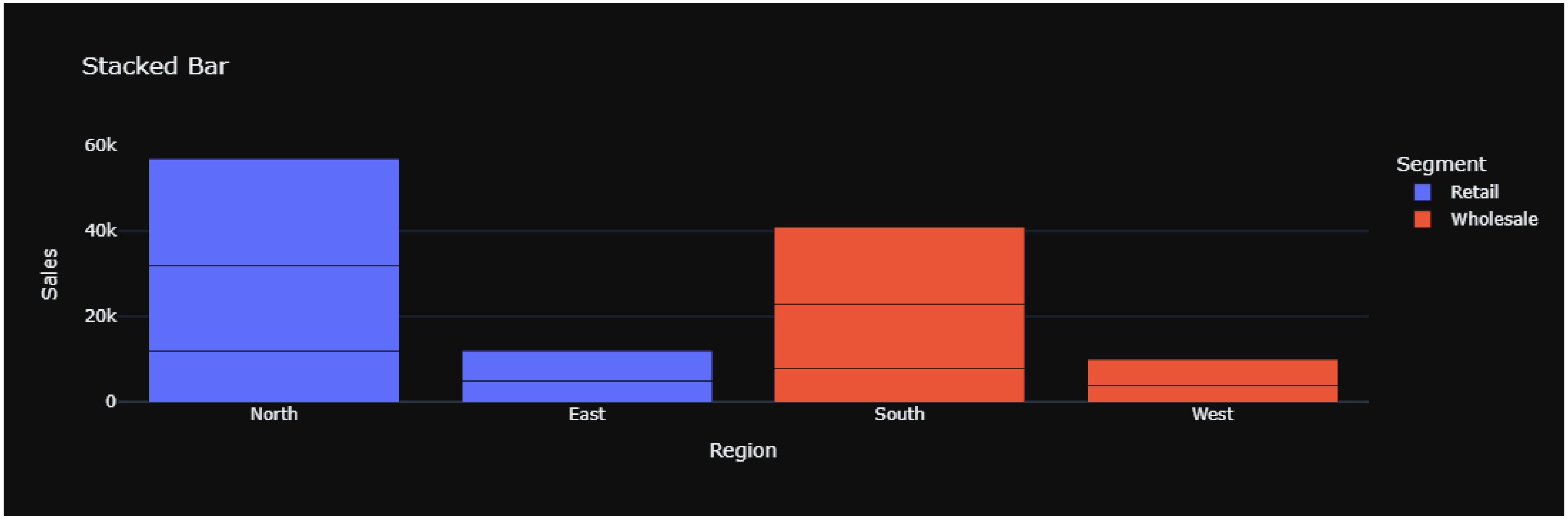
In [39]:
```
# grouped bar
BAZ.chart(df, "bar", x="Region", y="Sales", color="Segment",
```

```
         barmode="group", title="Grouped Bar")
```



In [38]:

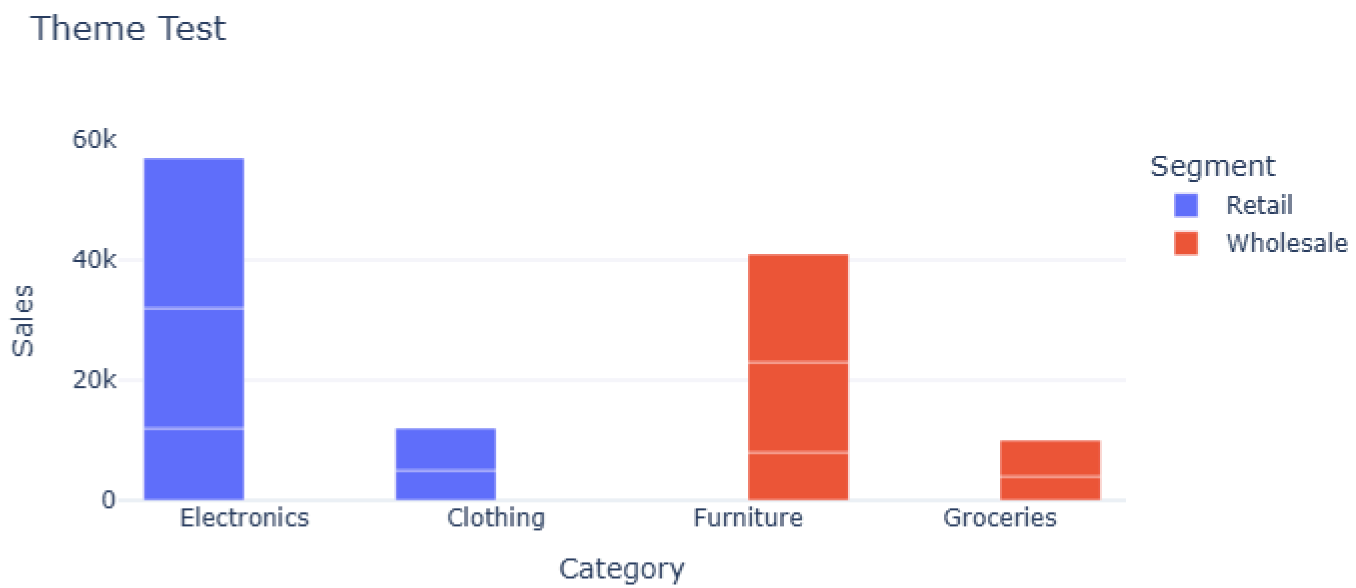
```
# stacked bar
BAZ.chart(df, "bar", x="Region", y="Sales", color="Segment",
         barmode="stack", title="Stacked Bar")
```



In [34]:

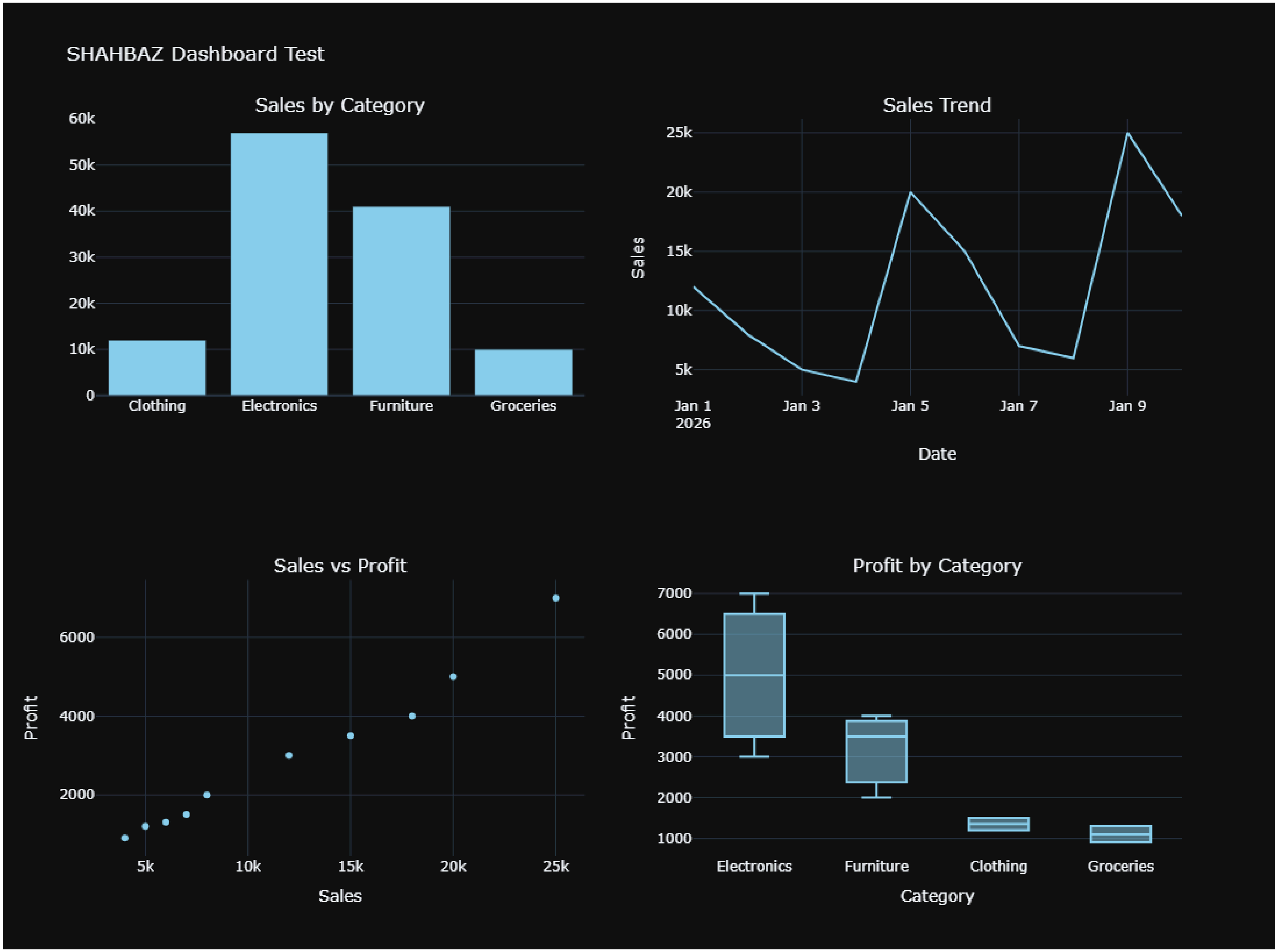
```
# THEME / COLOR TEST
```

In [35]:
```python
BAZ.chart(df, "bar", x="Category", y="Sales",
          title="Theme Test", theme="plotly_white", color="Segment")
```



In [71]:
```python
# DASHBOARD TEST
```

In [70]:
```python
fig = BAZ.dashboard([
    {"df": df, "chart_type": "bar", "groupby_col": "Category", "agg_col": "Sales", "agg_func": "sum", "title": "Sales
    {"df": df, "chart_type": "line", "x": "Date", "y": "Sales", "title": "Sales Trend"},
    {"df": df, "chart_type": "scatter", "x": "Sales", "y": "Profit", "title": "Sales vs Profit"},
    {"df": df, "chart_type": "box", "x": "Category", "y": "Profit", "title": "Profit by Category"}
], title="SHAHBAZ Dashboard Test", cols=2)

fig.show()
```


SHAHBAZ Dashboard Test
Sales by Category
60k
50k
40k
30k
20k
10k
0
Clothing
Electronics
Furniture
Groceries
Sales Trend
25k
20k
15k
10k
5k
Sales
Jan 1
2026
Jan 3
Jan 5
Jan 7
Jan 9
Date
Sales vs Profit
6000
4000
2000
Profit
5k
10k
15k
20k
25k
Sales
Profit by Category
7000
6000
5000
4000
3000
2000
1000
Profit
Electronics
Furniture
Clothing
Groceries
Category